

Skill Builder Feedback Report

Position: Skill Assessment - AI Engineer

Date: 6/5/2026

Session ID: dlai_interview_1e54582a4259411fbf660aa87af51e23



Skill Builder Overview

SUMMARY

The session focused on the development of a custom OCR and document intelligence pipeline for S'gaw Karen, a low-resource Southeast Asian language. You described building an end-to-end system involving synthetic data generation, YOLOv8 model training on rented GPUs, and a rule-based linguistic segmentation engine with LLM fallbacks.

STANDOUT STRENGTHS

Your technical grit in solving the 'cold start' problem for underrepresented languages is impressive. Specifically, you generated 27,000 synthetic syllable combinations to train a custom computer vision model that achieved an 88.7 mAP, demonstrating a deep understanding of the full machine learning lifecycle.

KEY DEVELOPMENT AREAS

There is an opportunity to move from manual, rule-based segmentation to statistical sub-word tokenization and to implement automated evaluation frameworks to replace manual verification of translation accuracy.

RECOMMENDED FOCUS

Develop an automated 'LLM-as-a-judge' evaluation pipeline to systematically compare your dictionary-matching results against multi-model fallbacks, reducing the need for manual linguistic review.



Recommended Learning Path

Curated based on your skill gap analysis



Document AI: From OCR to Agentic Doc Extraction

Short Course

This course directly aligns with your work on custom OCR and will help you transition from character recognition to structured document extraction using agentic pipelines.



Pydantic for LLM Workflows

Short Course

Since you are building complex Python-based pipelines and prompt generators, Pydantic will help you ensure data integrity and structured outputs across your dictionary and translation workflows.



DSPy: Build and Optimize Agentic Apps

Short Course

This will help you automate the 'vibe coding' and manual prompt refinement you described by using a programmatic approach to optimize your LLM interactions.





Evaluating AI Agents

Short Course

This course addresses your goal of creating a 'multi-model review' system to verify translation accuracy and handle hallucinations in your Gemini fallback mechanism.





Skill Assessments

Computer Vision & OCR Pipeline Development

Key Evidence:

- Trained a custom YOLOv8 model from scratch for S'gaw Karen character recognition.
- Generated 27,000 synthetic data samples covering every possible syllable combination in the script.
- Utilized Vast AI for GPU compute and RoboFlow for dataset management and annotation.
- Implemented preprocessing steps including noise reduction, cropping, tilting, and grayscale.

Path to Next Level:

To improve sequence-level accuracy, explore Transformer-based OCR architectures like TrOCR which can better capture the context of characters in a line. You can further enhance your synthetic data by using 'copy-paste' augmentation to place your generated syllables onto diverse, textured backgrounds that mimic real-world document aging. Finally, implement a sliding window inference approach to handle the 'dense region' problem you mentioned, ensuring high-resolution crops are fed to the model for complex script areas.

Computational Linguistics & Script Segmentation

Key Evidence:

- Developed a rule-based extraction method to handle 'no-space' script segmentation.
- Created a logic flow that uses grammatical particles and prepositions as anchors for word boundaries.
- Built a dictionary processing pipeline that scrapes and compares multiple sources to audit OCR output.
- Implemented a fallback mechanism using Gemini to recover cases where direct dictionary matching fails.

Path to Next Level:

Transition from manual rule-based segmentation to statistical methods using the SentencePiece library for sub-word tokenization, which is specifically designed for languages without clear word boundaries. Integrate a reranking step after your dictionary lookup that uses a cross-encoder to score the most likely translation based on the surrounding sentence context. This will help resolve ambiguities in your current 'first-match' iteration logic.

Prompt Engineering & Workflow Automation

Key Evidence:

- Built a custom prompt generator that refines user inputs based on roles and parameters.
- Designed a 'prompt pipeline' that suggests project outlines and generates sequential prompts for next steps.
- Automated the workflow for dataset generation and testing using approximately 40 Python files.

Path to Next Level:

To make your prompt pipeline more robust, use Pydantic to enforce structured outputs from your LLM calls, ensuring that refined prompts always follow a valid schema. You should also explore DSPy to programmatically optimize your prompts by treating them as code that can be tuned against a small set of 'gold' translation examples. This would replace manual prompt iteration with an automated optimization loop.